Murtaza Majid

Data Science Undergraduate

+92-311-5747189 | Murtazamajid.123@gmail.com | GitHub | LinkedIn | Portfolio

Summary

Data Science undergraduate at Air University with hands-on experience in machine learning, computer vision, and time-series forecasting, delivering production-style AI solutions across sports analytics, financial risk modeling, and real-time traffic monitoring.

Skills

Languages: Python, SQL
Data Science & ML: Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, LSTM, ARIMA, CNNs, NLP, OpenCV, MediaPipe, YOLOv11, Roboflow
Tools & Platforms: Power BI, MySQL, Streamlit, FastAPI, Supabase, Apache Spark, Microsoft Excel
Concepts: Time-Series Forecasting, Feature Engineering, Model Explainability (SHAP), Data Visualization

Experience

Computer Vision & AI Intern **Dec 2025 – Feb 2026**
Scoutrex, Islamabad *PyTorch, OpenCV, MediaPipe, Roboflow*
– Optimized **PyTorch** models for cricket video analysis under motion blur and varied lighting; implemented **MediaPipe** pose extraction for biomechanical analysis and bowler speed estimation via FPS-normalized displacement.
– Engineered an action phase segmentation pipeline to classify bowling stages (run-up to follow-through) using temporal motion tracking and multi-angle camera synchronization.

Data Science Intern **July 2025 – September 2025**
ITSOLERA Pvt. Ltd., Islamabad *Tools Used: Python, Pandas, Scikit-Learn, Streamlit, Plotly*
– Improved time-series forecast accuracy by **15%** using LSTM, ARIMA, XGBoost, and Random Forest models on business metrics.
– Built **interactive Streamlit dashboards** to visualize trends, KPIs, and model predictions for stakeholders.
– Developed end-to-end ML pipelines using Pandas and Scikit-Learn for real-world structured datasets.

LLM Task Trainer **[Dates]**
Turing *Prompt Engineering, RLHF, Tool Use*
– Engineered **system/user prompts** and **multi-step task chains** to optimize instruction following and integrate external API tools like Search, News, and Yelp using temporal logic.
– Mitigated **hallucinations** and improved model reasoning through rigorous **RLHF** evaluation and feedback, ensuring high-accuracy retrieval and context-aware response generation.

Content Writer **Jul 2022 – Aug 2022**
Xerosector Technologies *WordPress, SEO, Keyword Research*
– Created **50+ SEO-optimized** product descriptions, boosting organic traffic by **15%** through keyword optimization and brand strategy alignment.

Projects

AI-Based Traffic Violation Detection System (AutoCop) | *YOLOv11, OpenCV, FastAPI, PostgreSQL, Next.js*
– Developed a real-time AI traffic monitoring system detecting **6+ violation types** at **95%+ accuracy** with automated Gmail alerts for flagged events.
– Built a full-stack dashboard with live feeds, logs, analytics, and role-based access; deployed a scalable FastAPI backend with PostgreSQL storage. *(Final Year Project, Ongoing)*

Gmail Email Classification Pipeline | *Python, Scikit-Learn, TensorFlow, FastAPI, Supabase, Render, Vercel*
– Built an end-to-end NLP pipeline classifying **5,000+ Gmail emails** into five categories using a **13-step preprocessing pipeline** and TF-IDF; achieved up to **81.0% accuracy** with LSTM, outperforming Naive Bayes (72.6%) and Linear SVM (79.5%).
– Deployed a production-ready system with **FastAPI on Render**, integrating Gmail OAuth and Supabase, and delivered a live app on **Vercel** with model selection, batch inference, and per-class confidence scoring.

Awards & Certifications

Winner – Air MLX Hackathon 2025: Achieved **RMSE of 0.13** in a parking violation risk prediction system.

Professional Certifications: Intermediate Python (DataCamp); **Hadoop and Spark** (Coursera), focused on data structures, big data processing, and visualization.

Education

Bachelor of Science in Data Science **Expected: June 2026**
Air University, Islamabad